

Soumyanil Das

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EDUCATION

Haldia Institute of Technology

Bachelor of Technology in Information Technology

CGPA - 9.08

Sep '20 – Jun '24

EXPERIENCE

Software Engineer

Flutter, Dart, Python

[LightHearted AI](#)

Aug '24 – Present

- Architected cross-platform ecosystem encompassing Flutter mobile application, Linux desktop applications, and edge device firmware, achieving integration across 3+ platforms with zero data loss during inter-device communication
- Built dual-protocol communication stack supporting Bluetooth Low Energy (BLE) and USB serial transmission, delivering real-time PCG data transfer with < 50ms latency between medical devices and applications
- Developed edge computing solution including Python firmware modules with asynchronous processing pipeline, reducing camera data capture-to-transmission time from 7 minutes to under 7 seconds (98.3% improvement)
- Created manufacturing automation system featuring custom Linux OS flasher, automated firmware installation, and cloud synchronization pipeline, cutting device provisioning time by 85% and eliminating manual configuration errors
- Implemented FDA-compliant security layers with custom encryption and encapsulation over BLE protocol, maintaining data integrity for CE marking compliance and FDA approval readiness
- Accelerated signal processing algorithms achieving 7-second processing time on Mac and sub-20-second performance on Raspberry Pi, enabling deployment on low-power edge devices
- Designed custom USB serial communication protocol for desktop-to-edge device data transmission, maintaining 99.9% packet delivery rate with automatic error recovery

SDE Intern

Django, NextJs, AWS, Azure, Python, Networking, Scripting

[Fourie.ai](#)

Mar '24 – Jul '24

- Reduced audio wave sample bit rate to generate SVG files resulting in 27% drop in frontend memory usage.
- Developed a custom Admin Panel for asynchronous database modification tasks using Next.js and Django
- Implemented an OCR API using Tesseract with optimized settings, improving text recognition accuracy by 15%
- Implemented clipping of original audio segments to align with changed timestamps on S3
- Reduced the music source separation model's latency by 30% using Celery workers in a background job queue for parallel processing
- Written numerous scripts to perform data cleaning as well as automated tasks

Backend Developer Intern

Nodejs, Restful APIs, PostgreSQL, Typescript, HonoJs

[RecursiveZero](#)

Dec '23 – Mar '24

- Developed a custom API from scratch using TypeScript and Node.js
- Utilized PostgreSQL for database management, including writing custom SQL queries
- Deployed and managed the API on an AWS EC2 instance

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, GoLang

Frameworks: React, Node.js, Django, Gin, NextJs

Developer Tools: Git, VS Code, Visual Studio, Vim, AWS, Azure

Libraries: termios, Celery, gunicorn

PROJECTS

Video Conference Application | Nodejs, WebRTC, Sockets, Express

[Link](#)

- Developed a real-time video calling application to facilitate instant video communication between users
- Implemented a peer-to-peer connection using the PeerJS library, enabling secure and efficient data transfer between clients
- Created different rooms for meetings using web sockets by sockets.io
- Designed a dynamic URL generation system to create and join video conferencing rooms, allowing users to seamlessly enter video calls with unique URLs

vControl | Python, hashlib, zlib

[Link](#)

- Developed a fully functional version control system based on Git's architecture
- Used hashlib to use SHA-1 to compute hashes used in file and folder names
- Implemented major Git commands including checkout, init, commit etc.